

## Presentation 1

- Links to presentation(s) and code(s) on GitHub

- Presentation:  
[https://github.com/arvindkrishna87/STAT390\\_LegalAid\\_Fall2025/blob/main/Presentations/Time%20Trends%20analysis/TimeTrends\\_9Oct\\_Jasmine.pdf](https://github.com/arvindkrishna87/STAT390_LegalAid_Fall2025/blob/main/Presentations/Time%20Trends%20analysis/TimeTrends_9Oct_Jasmine.pdf)
- PowerBI Dashboard:  
<https://app.powerbi.com/view?r=eyJrIjoiazDMYnZc4ZjQtMmE3Ny00YjVjLWJlOWYtNzY3Yjg2YzJkYzg2Zm9udC16IjdhZkNzZkMzYxLTgyNzctNDcwOC1hNDc3LTY0ZTgzNjZjZDFiYyIsImMiOiN9>
- Code:  
<https://colab.research.google.com/drive/1ey2SEc4Tqeqscitl0wzuhuyI-aX3NPp?usp=sharing>

- What did you do?

- Explored the average call duration using AllCallsData, visualizing how the durations of both inbound and outbound calls vary throughout hours in a day and throughout days in a week.
- Proposed two metrics:
  - **Abandonment Rates:** using the “Answered” column from AllCallsData, the formula should look like  $\text{Calls with “Answered” = False} / \text{total inbound calls}$ . This new metric will help quantify missed opportunities for service and helps justify resource increases or chatbot automation. By looking at how the abandonment rates fluctuate over different time periods, we can discover when there’s a lack in staffing for LegalAid.
  - **Repeat Caller Ratio:** using the “Calling number” column from AllCallsData, we can identify the number of calling numbers with more than 1 inbound call / Total calls, which can help us identify if users are not getting help during their first calls. Ideally, the smaller the repeat caller ratio, the better LegalAid serves its callers.

- How does it help the project?

The average call duration in a particular time period is an important metric for improving LegalAid’s phone call system, since it provides an indirect measure of the complexity or the difficulty of the tasks handled by the calls. By looking at how the average call duration changes over time, we can understand both LegalAid’s operational schedule and callers’ behavior. For example, if the average call duration for inbound calls in the afternoon is longer than that in the morning, it indicates that callers tend to inquire about more complex issues in the afternoon. Similarly, if the average call duration for outbound calls in work days is longer than that at weekends, it could

suggest that LegalAid staff tend to tackle more time-consuming issues through Monday to Friday. We can discover whether there's misalignment between the organization's schedule and callers' expectations by examining the average call duration of inbound and outbound calls.

Based on my visualization for the average call duration throughout hours in a day, I found that the average duration for both inbound and outbound calls stay relatively similar from 5AM to 5PM. However, after 5PM, the average duration of outbound calls reduced significantly, while that of inbound calls increased dramatically and reached its peak between 8PM and 10PM. This divergence indicates that, while the call operators might be winding down their operations and tend to address simpler issues after normal work hours, the callers are actually calling to deal with more complex or more time-consuming tasks. The pattern of average call duration suggests a possible misalignment between the organization's operational hours and the public's needs, implying that resource allocation or staffing during evening hours may not fully align with when callers expect support for more time-consuming issues.

For the visualization of average call duration throughout days in a week, both the inbound and outbound duration remains relatively the same from Tuesday to Sunday, yet the duration of inbound calls is about 1-minute longer than the duration of outbound calls on Mondays, which might suggest that the callers tend to inquire about more time-consuming issues at the start of a week yet the call operators prefer to handle simpler issues reflected by the shorter duration of outbound calls.

To better align LegalAid's operations with caller needs, the organization could consider extending or reallocating staff coverage during evening hours, when callers tend to bring up more complex issues. Additionally, scheduling more experienced staff or specialized training sessions earlier in the week—particularly on Mondays—may help address the surge in longer inbound calls. Regularly monitoring call duration patterns by time of day and day of week can further guide data-driven adjustments in staffing and scheduling to improve overall service efficiency and caller satisfaction.

- Issues faced (if any)

While I initially used the "Direction" column to determine if a call is inbound or outbound, this column alone cannot show if the calls are truly inbound or outbound, since it only indicates whether a call originates or terminates at LegalAid. I need to also take the "PSTN vendor name" column into consideration, looking at the values for both columns and removing the internal calls to avoid confusion.

- Attempts to resolve issues (if any)

To distinguish between inbound and outbound calls, I first sorted the observations in AllCallsData by "Correlation ID" and "Start time," then retained the first record for each unique correlation ID. Calls were categorized based on the "PSTN vendor name" and

“Direction” fields: if the “PSTN vendor name” was “CallTower” and the “Direction” was “Terminating,” the call was classified as inbound; if “CallTower” and “Originating,” it was classified as outbound. When “PSTN vendor name” was “NA,” and “Direction” indicated either “Terminating” or “Originating,” the call was identified as internal. Following this classification, I aggregated the “Duration” values for all rows sharing the same correlation ID to accurately represent the total duration of each call before re-visualizing the average call duration.

- Issues resolved (if any)

By preprocessing the data through methods mentioned above, I was able to correctly visualize the average call duration for inbound and outbound calls.

- Next steps

Assuming that the inbound and outbound calls have been correctly identified, I plan to investigate average call duration in the following directions:

- Identify if specific phone menus or call/client types are associated with the time periods showing higher average call duration
- Examine caller wait times during periods with higher average call durations to assess potential understaffing during peak hours
- Evaluate whether call timing influences case outcomes
- Assess if periods with long average call duration are seasonal or constantly recurring

- References (Mention if you built up on someone else’s work)